

Supplementary Information: Exact architecture exploration for heterogeneous edge and cloud cyber-physical systems

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Abstract

Supplementary Notes 1–2 and Supplementary Tables S1–S13 accompanying the main manuscript. All tabulated values are generated from the same result files as the main-text summaries.

1 Supplementary Note 1: correctness of the pruning rules

1.1 Setting

The design space is the Cartesian product $\mathcal{X} = D_1 \times \dots \times D_k$ traversed under a fixed variable order $v_1, \dots, v_k$. A *partial design* of depth d is a prefix of assignments $p = (a_1, \dots, a_d)$, $a_i \in D_{v_i}$, and its completions are $\mathcal{C}(p) = \{x \in \mathcal{X} : x_{v_i} = a_i, i \leq d\}$.

Let $\sigma : \mathcal{X} \rightarrow \{0, 1\}$ be the well-formedness predicate, $\mathcal{X}_\sigma$ the well-formed space, $\mathcal{F}$ the feasible set and $\mathcal{F}_\sigma = \mathcal{F} \cap \mathcal{X}_\sigma$. Objectives $F(x) = (f_1(x), \dots, f_n(x))$ are minimised, $u \preceq v$ denotes $u_i \leq v_i$ for all i with at least one strict inequality, and

$$\mathcal{P}^* = \{F(x) : x \in \mathcal{F}_\sigma, \nexists y \in \mathcal{F}_\sigma : F(y) \preceq F(x)\}. \quad (\text{S1})$$

A bound LB_i is *admissible* for f_i if $LB_i(p) \leq f_i(x)$ for every partial design p and every $x \in \mathcal{C}(p)$.

1.2 Construction of the bounds

Supplementary Theorem 1 (Term-wise relaxation) *Let $f_i(x) = \sum_{t \in T_i} \varphi_t(x)$ with $\varphi_t \geq 0$ and $\underline{\varphi}_t(p) = \min_{x \in \mathcal{C}(p)} \varphi_t(x)$. Then $LB_i(p) = \sum_{t \in T_i} \underline{\varphi}_t(p)$ is admissible.*

Proof For every $x \in \mathcal{C}(p)$ and every $t \in T_i$ we have $\underline{\varphi}_t(p) \leq \varphi_t(x)$, because the minimum is taken over a set that contains x . Summing over $t \in T_i$ gives $LB_i(p) \leq \sum_t \varphi_t(x) = f_i(x)$. $\square$

The minima are taken independently per term, so the combination of values realising them need not correspond to any single completion; the bound is optimistic but valid. Because the decision domains are finite, the minimum of each term over the reachable domain is precomputed exactly and read from a table in constant time; for ordered scalar domains the minimum is attained at a domain endpoint, which is how the table is built.

Supplementary Theorem 2 (Exactness on determined terms) *Let $\text{supp}(\varphi_t)$ be the set of variables φ_t depends on. If $\text{supp}(\varphi_t) \subseteq \{v_1, \dots, v_d\}$, then φ_t is constant on $\mathcal{C}(p)$, and substituting its exact value for $\underline{\varphi}_t(p)$ preserves admissibility while tightening the bound.*

Proof Constancy follows from the definition of $\mathcal{C}(p)$, and a constant equals its own minimum over $\mathcal{C}(p)$, so Theorem 1 applies with the substituted value. $\square$

Remark 1 (Role of the variable order) The order places the determinants of tier utilisation — node counts, node classes, replication factor and placement policy — before the remaining variables. Once they are assigned, the waiting term $\frac{w_i}{C_t} \cdot \frac{\rho_t}{2(1-\rho_t)}$ is constant on $\mathcal{C}(p)$ and enters the bound exactly by Theorem 2 instead of being relaxed to its trivial bound 0. This is what makes a contention-dependent objective usable for safe pruning without assignment monotonicity. If the determinants are fixed and $\rho_t \geq 1$, no completion has finite latency and $LB_L(p) = \infty$, which is admissible.

1.3 The three pruning rules

Supplementary Proposition 3 (Safe feasibility pruning) *If LB_{g_j} is admissible for the constrained quantity g_j and $LB_{g_j}(p) > g_j^{\max}$, then $\mathcal{C}(p) \cap \mathcal{F} = \emptyset$.*

Proof For any $x \in \mathcal{C}(p)$, admissibility gives $g_j(x) \geq LB_{g_j}(p) > g_j^{\max}$, so x violates the constraint and $x \notin \mathcal{F}$. $\square$

Supplementary Proposition 4 (Safe dominance pruning) *Let $y \in \mathcal{F}_\sigma$ be a fully evaluated feasible architecture with $F(y) \preceq LB(p)$. Then $F(x) \notin \mathcal{P}^*$ for every $x \in \mathcal{C}(p)$.*

Proof Let $x \in \mathcal{C}(p)$. Admissibility gives $f_i(x) \geq LB_i(p) \geq F_i(y)$ for all i , and by assumption there is a j with $F_j(y) < LB_j(p) \leq f_j(x)$. Hence y strictly dominates x , so $F(x)$ is not on the Pareto front. $\square$

Supplementary Proposition 5 (Closure of the well-formedness predicate) *Each invariant of σ depends only on assigned variables and, under the chosen order, all variables of an invariant are assigned before it is first checked. Since assignments are never revised, $\neg\sigma(p)$ implies $\sigma(x) = 0$ for all $x \in \mathcal{C}(p)$, that is $\mathcal{C}(p) \cap \mathcal{X}_\sigma = \emptyset$.*

Remark 2 The structural invariants define what counts as an architecture rather than following from the constraints of the problem: the evaluator can formally compute objectives for, say, a replicated deployment under a policy that places the ingest stage off the edge tier, although such a configuration is meaningless. Structural pruning is therefore part of the definition of the search space, and the exhaustive baseline applies the same predicate.

1.4 Front preservation

Supplementary Theorem 6 (Pareto-front preservation) *If all bounds are admissible, pruning is performed only by Propositions 3–5, and the archive contains only feasible fully evaluated architectures, then the set of non-dominated vectors returned by the exploration equals $\mathcal{P}^*$.*

Proof ($\subseteq$) Every returned vector is $F(x)$ for an evaluated $x \in \mathcal{F}_\sigma$ that survives non-dominated filtering. A vector dominating it would belong to an architecture that was either evaluated, in which case filtering would have removed the dominated one, or pruned; pruning by Proposition 3 or 5 would contradict its feasibility and well-formedness, and pruning by Proposition 4 implies a further feasible evaluated architecture dominating it, so the argument repeats along a strictly decreasing chain in a finite set.

($\supseteq$) Let $u \in \mathcal{P}^*$ and $x \in \mathcal{F}_\sigma$ with $F(x) = u$. The traversal is a prefix tree, so x is reached through the unique chain $p_0 \subset p_1 \subset \dots \subset p_k = x$. If some p_d were pruned, then by Proposition 5 $\sigma(x) = 0$ and by Proposition 3 $x \notin \mathcal{F}$, both contradicting $x \in \mathcal{F}_\sigma$; by Proposition 4 there would exist a feasible y strictly dominating x , contradicting $u \in \mathcal{P}^*$. Hence no p_d is pruned, x is evaluated, and u survives the final filtering. $\square$

Remark 3 (Duplicates) The theorem is stated in objective space. If two distinct architectures share an objective vector, dominance pruning may retain only one of them; the front is unaffected. All experiments therefore compare sets of objective vectors.

Remark 4 (Order independence) Proposition 4 holds for any archive of evaluated feasible architectures, so the traversal order affects pruning efficiency but never correctness. This is confirmed empirically in Supplementary Table S7, where all order and toggle combinations return the same front.

1.5 Complexity and break-even

With C_E the cost of one full evaluation, C_P the cost of computing the bounds at one node, N the number of expanded partial designs, N_E the number of full evaluations and $|A|$ the archive size,

$$T_{\text{exh}} = O(|\mathcal{X}| C_E), \quad T_{\text{HCA}} = O(N(C_P + n|A|) + N_E C_E), \quad (\text{S2})$$

so pruning is profitable when $C_E > N(C_P + n|A|)/(|\mathcal{X}| - N_E)$. The right-hand side is measured rather than assumed. In the runs reported here it is negative on every scale and workload, because the total search overhead $N(C_P + n|A|)$ is already smaller than the overhead of enumerating the space: there is no positive evaluation cost below which exhaustive enumeration wins, and the advantage grows from there with C_E .

2 Supplementary Note 2: full evaluation model

With $\mathcal{H}(x)$ the hop set implied by the placement policy and the replication factor, Λ the system arrival rate, T the accounting horizon and ρ_t the tier utilisation defined in Methods:

$$E(x) = \sum_{t \in \{e, g\}} N_t \left(P_t^{\text{idle}} + \rho_t (P_t^{\text{peak}} - P_t^{\text{idle}}) \right) T + \rho_c P_c^{\text{peak}} T + \sum_{(s, \ell) \in \mathcal{H}(x)} s \Lambda e_\ell T, \quad (\text{S3})$$

$$C(x) = n_e (c_{\kappa_e} + c_\ell) + n_g (c_{\kappa_g} + c_\ell) + c_{\kappa_c}, \quad (\text{S4})$$

$$D(x) = \Lambda \sum_{(s, \ell) \in \mathcal{H}(x)} s. \quad (\text{S5})$$

An architecture is feasible when $\rho_t \leq 0.85$ on every used tier, the resident memory of the stages mapped to a tier fits the memory of its node class, the aggregate access-link traffic does not exceed the total bandwidth of the attached nodes, and the latency, cost and energy budgets hold. Energy is reported in kWh over $T = 1$ h. Cost coefficients are synthetic normalised values encoding relative price tiers; no monetary interpretation is claimed, and neither the energy nor the cost model is validated against measurements.

3 Supplementary Tables

The main text compares analytical predictions with the mean latency measured over each deployment run. Supplementary Table S4 is the complementary robustness analysis, in which the same predictions are compared with the measured median (p_{50}) of the same runs.

Table S1 Full budget-matched comparison on S2. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S2	telemetry	exhaustive	3 888	1.000	1.000	0.0000	0.045
		random ($B=100$)	100	0.037	0.692	0.1496	0.001
		random ($B=250$)	250	0.037	0.794	0.1018	0.003
		random ($B=500$)	500	0.074	0.880	0.0713	0.007
		random ($B=1000$)	1000	0.185	0.923	0.0451	0.014
		NSGA-II ($B=100$)	100	0.037	0.703	0.1537	0.011
		NSGA-II ($B=250$)	250	0.259	0.886	0.0447	0.029
		NSGA-II ($B=500$)	500	0.759	0.995	0.0063	0.072
		NSGA-II ($B=1000$)	1000	0.944	0.999	0.0003	0.148
		HCA-DSE	46	1.000	1.000	0.0000	0.013
	control	exhaustive	3 888	1.000	1.000	0.0000	0.049
		random ($B=100$)	100	0.040	0.691	0.1460	0.001
		random ($B=250$)	250	0.040	0.792	0.0940	0.003
		random ($B=500$)	500	0.080	0.879	0.0566	0.006
		random ($B=1000$)	1000	0.160	0.923	0.0379	0.014
		NSGA-II ($B=100$)	100	0.040	0.627	0.1592	0.011
		NSGA-II ($B=250$)	250	0.300	0.910	0.0348	0.030
		NSGA-II ($B=500$)	500	0.800	0.991	0.0041	0.072
		NSGA-II ($B=1000$)	1000	0.900	0.999	0.0007	0.161
		HCA-DSE	39	1.000	1.000	0.0000	0.014
S2	sensing	exhaustive	3 888	1.000	1.000	0.0000	0.029
		random ($B=100$)	100	0.000	0.469	0.2629	0.001
		random ($B=250$)	250	0.062	0.686	0.1521	0.002
		random ($B=500$)	500	0.062	0.804	0.0808	0.005
		random ($B=1000$)	1000	0.188	0.866	0.0622	0.010
		NSGA-II ($B=100$)	100	0.000	0.404	0.3482	0.010
		NSGA-II ($B=250$)	250	0.312	0.793	0.0782	0.031
		NSGA-II ($B=500$)	500	0.719	0.965	0.0224	0.077
		NSGA-II ($B=1000$)	1000	0.938	1.000	0.0009	0.186
		HCA-DSE	26	1.000	1.000	0.0000	0.010

Table S2 Full budget-matched comparison on S3. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S3	telemetry	exhaustive	34 560	1.000	1.000	0.0000	0.320
		random ($B=100$)	100	0.000	0.259	0.3903	0.001
		random ($B=250$)	250	0.000	0.487	0.2574	0.002
		random ($B=500$)	500	0.000	0.563	0.1671	0.005
		random ($B=1000$)	1000	0.016	0.699	0.1147	0.011
		NSGA-II ($B=100$)	100	0.000	0.151	0.4919	0.009
		NSGA-II ($B=250$)	250	0.000	0.687	0.1732	0.026
		NSGA-II ($B=500$)	500	0.355	0.915	0.0558	0.061
		NSGA-II ($B=1000$)	1000	0.645	0.987	0.0415	0.146
		HCA-DSE	52	1.000	1.000	0.0000	0.055
	control	exhaustive	34 560	1.000	1.000	0.0000	0.276
		random ($B=100$)	100	0.000	0.262	0.4192	0.001
		random ($B=250$)	250	0.000	0.492	0.2537	0.002
		random ($B=500$)	500	0.000	0.564	0.1746	0.005
		random ($B=1000$)	1000	0.018	0.691	0.1248	0.010
		NSGA-II ($B=100$)	100	0.000	0.286	0.4405	0.010
		NSGA-II ($B=250$)	250	0.037	0.748	0.1009	0.025
		NSGA-II ($B=500$)	500	0.278	0.933	0.0387	0.063
		NSGA-II ($B=1000$)	1000	0.648	0.991	0.0244	0.147
		HCA-DSE	43	1.000	1.000	0.0000	0.053
S3	sensing	exhaustive	34 560	1.000	1.000	0.0000	0.282
		random ($B=100$)	100	0.000	0.037	0.6082	0.001
		random ($B=250$)	250	0.000	0.231	0.3843	0.002
		random ($B=500$)	500	0.000	0.556	0.2543	0.005
		random ($B=1000$)	1000	0.000	0.600	0.1747	0.010
		NSGA-II ($B=100$)	100	0.000	0.050	0.5191	0.011
		NSGA-II ($B=250$)	250	0.037	0.279	0.2833	0.025
		NSGA-II ($B=500$)	500	0.370	0.750	0.1996	0.064
		NSGA-II ($B=1000$)	1000	0.741	0.969	0.1670	0.165
		HCA-DSE	43	1.000	1.000	0.0000	0.071

Table S3 Full S3 statistical comparison: one-sided Mann–Whitney test for larger NSGA-II hypervolume and Cliff’s δ . Unadjusted p -values, exploratory.

workload	budget	NSGA-II HV med.	random HV med.	p	Cliff’s δ	NSGA-II recall	HCA-DSE evals
telemetry	100	0.1509	0.259	0.951	-0.302	0.0	52
telemetry	250	0.6868	0.4871	0.021	0.38	0.0	52
telemetry	500	0.9154	0.5633	<0.001	0.7	0.355	52
telemetry	1000	0.9874	0.6993	<0.001	0.7	0.645	52
control	100	0.2859	0.2616	0.484	0.01	0.0	43
control	250	0.7481	0.4916	<0.001	0.62	0.037	43
control	500	0.9331	0.5644	<0.001	0.8	0.278	43
control	1000	0.9906	0.6913	<0.001	0.8	0.648	43
sensing	100	0.0497	0.0367	0.261	0.12	0.0	43
sensing	250	0.2787	0.2308	0.672	-0.08	0.037	43
sensing	500	0.7504	0.5561	0.168	0.18	0.37	43
sensing	1000	0.9689	0.5998	0.054	0.3	0.741	43

Table S4 Testbed validation against measured p_{50} , including utilisation and reproducibility subsets. Pair agreement is a fraction; undefined values (no eligible pairs) are shown as –.

subset	n	MAPE [%]	med. AE [ms]	Spearman ρ	Kendall τ	pair > 10%	pair > 25%	repeat IQR [ms]
telemetry (all)	12	19.9	7.43	0.811	0.697	0.815	0.895	0.23
telemetry ($\rho \leq 0.75$)	6	15.4	2.85	0.829	0.733	0.818	1.0	0.2
control (all)	12	14.4	2.49	0.865	0.748	0.896	0.909	0.11
control ($\rho \leq 0.75$)	10	9.4	2.32	0.766	0.629	0.815	0.833	0.11
sensing (all)	12	51.1	19.15	0.935	0.84	0.935	0.913	1.0
sensing ($\rho \leq 0.75$)	10	51.0	18.82	0.985	0.944	1.0	1.0	0.75
sensing (reproducible)	8	39.3	18.56	0.97	0.909	1.0	–	0.67
all workloads (all)	36	28.4	7.43	0.968	0.884	0.961	0.98	0.27
all workloads ($\rho \leq 0.75$)	26	26.8	3.96	0.945	0.855	0.959	0.982	0.22
all workloads (reproducible)	32	22.7	4.5	0.957	0.869	0.956	0.979	0.23

Table S5 Exact solver baselines against HCA-DSE on S1–S3. Finite-table SMT evaluates every structurally valid design and then solves; factorised SMT tabulates each model term over its own decisions only and calls no evaluator. “Evaluator calls” counts full design-point evaluations; “tabulated entries” counts table rows built before solving; total time includes table construction. All rows return the exhaustive front. Factorised SMT requires the complete analytical objective and constraint model to be representable by the factor tables; solver times depend on the encoding.

scale	workload	method	evaluator calls	tabulated entries	solver checks	total [s]	front
S1	telemetry	HCA-DSE	32	–	–	0.005	17
		finite-table SMT	1080	1080	44	0.341	17
		factorised SMT	0	221	51	1.518	17
S1	control	HCA-DSE	28	–	–	0.005	18
		finite-table SMT	1080	1080	48	0.317	18
		factorised SMT	0	225	52	1.684	18
S1	sensing	HCA-DSE	16	–	–	0.004	11
		finite-table SMT	1080	1080	31	0.155	11
		factorised SMT	0	199	37	0.766	11
S2	telemetry	HCA-DSE	46	–	–	0.014	27
		finite-table SMT	3888	3888	71	3.305	27
		factorised SMT	0	529	81	10.613	27
S2	control	HCA-DSE	39	–	–	0.013	25
		finite-table SMT	3888	3888	61	2.180	25
		factorised SMT	0	537	83	12.447	25
S2	sensing	HCA-DSE	26	–	–	0.011	16
		finite-table SMT	3888	3888	48	0.937	16
		factorised SMT	0	469	50	7.373	16
S3	telemetry	HCA-DSE	52	–	–	0.058	31
		finite-table SMT	0	2384	87	80.961	31
		factorised SMT	43	–	–	0.055	27
S3	control	HCA-DSE	43	–	–	0.055	27
		finite-table SMT	0	2406	85	94.321	27
		factorised SMT	43	–	–	0.067	27
S3	sensing	HCA-DSE	43	–	–	0.067	27
		finite-table SMT	0	2188	85	71.363	27
		factorised SMT	0	–	–	–	–

Table S6 Ablation on S3: expanded partial designs, full evaluations and runtime for each workload.

variant	telemetry			control			sensing		
	states	evals	[ms]	states	evals	[ms]	states	evals	[ms]
structural only	61 225	34 560	369	61 225	34 560	284	61 225	34 560	284
structural + feasibility bounds	24 650	12 700	359	15 979	7 025	262	23 118	10 956	315
structural + dominance bounds	4 617	323	105	4 730	405	100	4 804	412	89
HCA-DSE (all layers)	2 397	52	54	2 375	43	51	3 221	43	61

Table S7 Full order and queue-bound ablation on S1–S3. All configurations preserve the unrounded exhaustive front.

scale	workload	order	queue	states	checks	evals	active	pruned	ms
S1	telemetry	determinants-early	off	447	746	136	0	0	6.7
S1	telemetry	determinants-early	on	249	516	32	306	46	5.2
S1	telemetry	determinants-late	off	695	1299	136	0	0	9.5
S1	telemetry	determinants-late	on	589	1299	30	0	0	10.3
S1	control	determinants-early	off	391	669	113	0	0	5.2
S1	control	determinants-early	on	242	512	28	312	35	4.6
S1	control	determinants-late	off	679	1317	112	0	0	9.6
S1	control	determinants-late	on	593	1317	26	0	0	10.5
S1	sensing	determinants-early	off	481	810	147	0	0	5.8
S1	sensing	determinants-early	on	226	503	16	309	72	4.1
S1	sensing	determinants-late	off	497	819	148	0	0	6.1
S1	sensing	determinants-late	on	365	819	16	0	0	5.8
S2	telemetry	determinants-early	off	939	1815	208	0	0	15.4
S2	telemetry	determinants-early	on	604	1383	46	839	99	13.9
S2	telemetry	determinants-late	off	1816	3940	208	0	0	30.7
S2	telemetry	determinants-late	on	1652	3940	44	0	0	35.8
S2	control	determinants-early	off	835	1674	165	0	0	13.9
S2	control	determinants-early	on	587	1364	39	842	80	12.0
S2	control	determinants-late	off	1780	3960	164	0	0	29.0
S2	control	determinants-late	on	1653	3960	37	0	0	32.9
S2	sensing	determinants-early	off	1161	2184	289	0	0	15.0
S2	sensing	determinants-early	on	600	1427	26	879	177	10.1
S2	sensing	determinants-late	off	1268	2421	290	0	0	16.8
S2	sensing	determinants-late	on	1004	2421	26	0	0	15.8
S3	telemetry	determinants-early	off	2871	6806	251	0	0	53.3
S3	telemetry	determinants-early	on	2397	6100	52	3777	188	54.4
S3	telemetry	determinants-late	off	6029	14794	251	0	0	117.9
S3	telemetry	determinants-late	on	5828	14794	50	0	0	131.9
S3	control	determinants-early	off	2755	6636	206	0	0	51.0
S3	control	determinants-early	on	2375	6074	43	3777	163	56.3
S3	control	determinants-late	off	5990	14812	205	0	0	106.7
S3	control	determinants-late	on	5826	14812	41	0	0	127.5
S3	sensing	determinants-early	off	4351	9905	537	0	0	72.9
S3	sensing	determinants-early	on	3221	8281	43	5139	419	60.9
S3	sensing	determinants-late	off	4888	11367	538	0	0	82.7
S3	sensing	determinants-late	on	4393	11367	43	0	0	82.3

Table S8 Design variables and domain sizes of the four design-space scales.

variable	meaning	S1	S2	S3	S4
<i>n_e</i>	number of edge nodes	3	6	24	80
<i>edge_cls</i>	edge node class	3	3	3	3
<i>n_g</i>	number of gateways	2	3	5	8
<i>gw_cls</i>	gateway class	2	2	2	2
<i>rep</i>	replication factor of stage 1	2	3	4	5
<i>policy</i>	placement policy	3	3	3	3
<i>link_cls</i>	access link class	3	3	3	3
<i>cloud_cls</i>	cloud instance class	2	2	2	2
design-space size $ \mathcal{X} $		1 296	5 832	51 840	345 600

Table S9 Workload classes. Stage values are given as $\tau_1/\tau_2/\tau_3$.

workload	n_d	λ	Λ	w_i [MI]	s_i [kB]	D [ms]	$C^{\max}$
telemetry	200	2	400	8/4/20	20/10/10	120	60
control	60	10	600	5/3/9	5/4/3	30	60
sensing	40	1.5	60	120/30/260	1200/350/300	900	90

Table S10 Resource classes used in all experiments.

class	C	B	M	τ	P^{idle}	e	P^{peak}	cost
<i>compute node classes</i>								
edge-low	2000		2048		4	12	1	
edge-med	6000		8192		8	30	2.4	
edge-high	15000		16384		15	65	5.2	
gw-low	4000		4096		6	20	1.8	
gw-high	12000		16384		12	48	4	
cloud-med	40000		131072		0	180	6	
cloud-high	100000		262144		0	380	12	
<i>link classes</i> (name, B [Mb/s], τ [ms], e [J/MB], cost)								
constrained	10		4		0.3	0.2	–	
standard	100		2		0.12	0.6	–	
fast	1000		1		0.05	1.4	–	

Table S11 Exploration funnel across all scales and workloads.

scale	workload	$ \mathcal{X} $	structural pruned	feasibility pruned	dominance pruned	full evals	feasible	$ \mathcal{P} $	full-eval reduction [%]	exhaustive evals	recall
S1	telemetry	1 296	216	208	840	32	25	17	97.53	1 080	1.0
S1	control	1 296	216	226	826	28	25	18	97.84	1 080	1.0
S1	sensing	1 296	216	530	534	16	11	11	98.77	1 080	1.0
S2	telemetry	5 832	1 860	965	2 961	46	38	27	99.21	3 888	1.0
S2	control	5 832	1 860	1 017	2 916	39	36	25	99.33	3 888	1.0
S2	sensing	5 832	1 944	1 534	2 328	26	19	16	99.55	3 888	1.0
S3	telemetry	51 840	12 240	21 957	17 591	52	44	31	99.90	34 560	1.0
S3	control	51 840	12 240	21 924	17 633	43	40	27	99.92	34 560	1.0
S3	sensing	51 840	14 634	16 230	20 933	43	36	27	99.92	34 560	1.0
S4	telemetry	345 600	33 924	276 742	34 877	57	49	32	99.98	–	–
S4	control	345 600	33 924	276 811	34 812	53	50	29	99.98	–	–
S4	sensing	345 600	41 250	230 523	73 775	52	45	28	99.98	–	–

Table S12 Scalability sweep: expanded partial designs, full evaluations and runtime.

$ \mathcal{X} $	telemetry			control			sensing		
	states	evals	[s]	states	evals	[s]	states	evals	[s]
1.3×10^3	286	42	0.05	268	34	0.05	234	18	0.04
5.8×10^3	696	54	0.14	682	54	0.13	649	30	0.10
5.2×10^4	2 746	57	0.60	2 732	57	0.52	3 495	53	0.65
3.5×10^5	5 092	58	1.14	5 074	58	1.05	8 228	59	1.63
2.8×10^6	8 025	58	1.88	8 007	58	1.70	13 986	61	3.01
1.6×10^7	9 252	58	2.22	9 234	58	2.04	16 345	61	3.67
8.3×10^7	10 333	58	2.13	10 315	58	2.50	18 082	61	3.96

Table S13 Analytical mean latency versus the discrete-event evaluator: 40 designs per workload, five seeds per design.

workload	designs	MAPE [%]	Spearman ρ
telemetry	40	6.9	0.986
control	40	6.1	0.988
sensing	40	20.5	0.955